Science, Mathematics and Technology

SSILP Math

Beginner

QUADRATIC FUNCTIONS & RATIONAL FUNCTIONS

Learning objectives

Understand the Basics of Polynomial and Rational Functions

Graphing and Analyzing Quadratic Functions

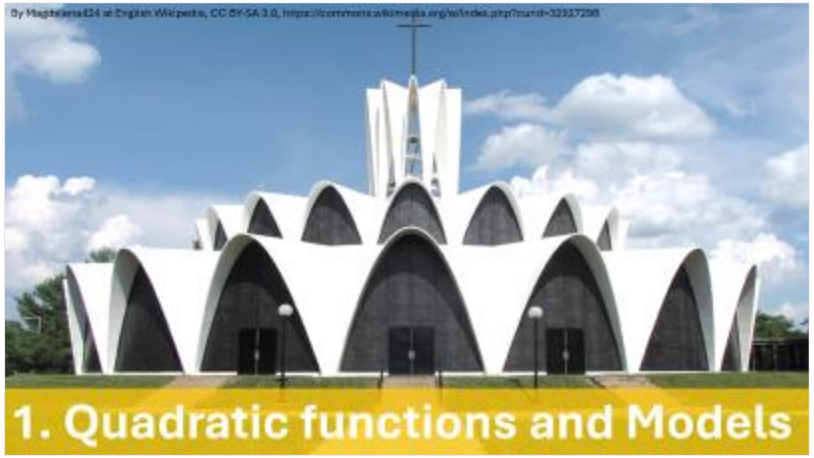
- Graph quadratic functions and identify key features such as the vertex, axis of symmetry, and intercepts
- Solve quadratic equations using different methods, including **factoring**, **completing the square**, and the **quadratic formula**

Understanding Rational Functions and Their Graphs

- Identify and graph rational functions, including understanding **vertical and horizontal asymptotes**
- Determine the **domain of rational functions**

Session outline

Chapter 3
Introduction:
Polynomial and Rational Functions



a. Quadratic Functions:
General Form

b. Quadratic functions:
Standard form

Refer to the textbook!

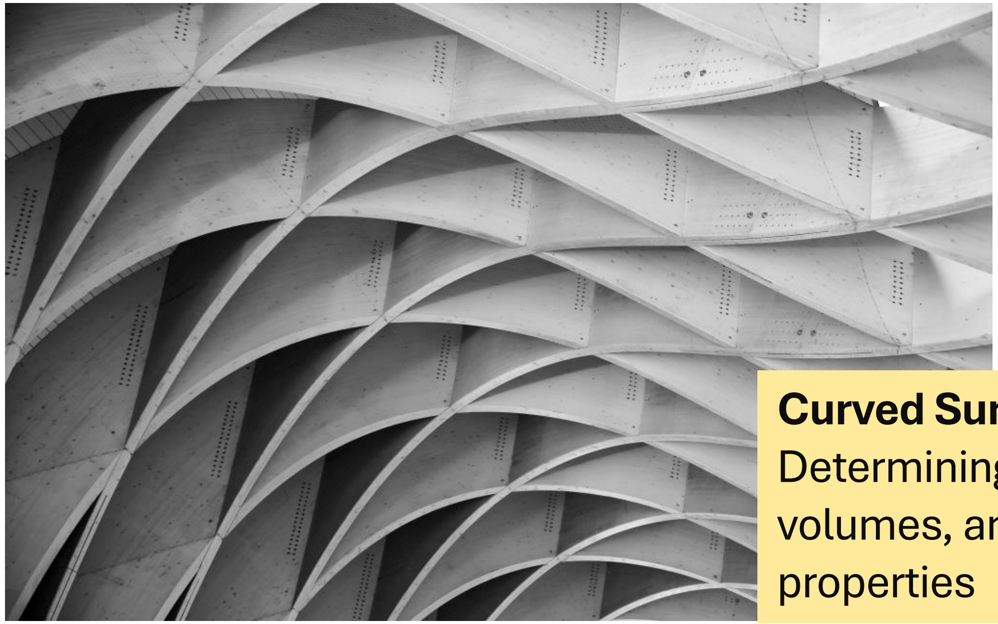
Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7^{th} Ed)

Chapter 3, Section 3.1, 3.2, 3.3

Chapter 3
Introduction:
Polynomial and Rational Functions

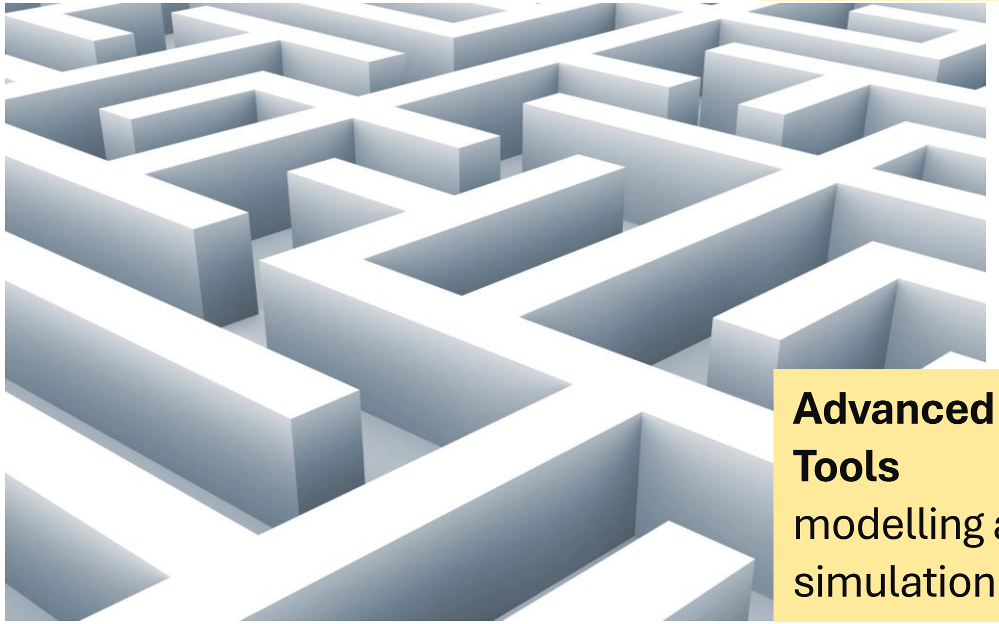
Prelude: Where are some applications of calculus?

Structural Analysis understanding the forces and moments acting on structures



Curved Surfaces: Determining areas, volumes, and other properties

Optimization for factors like cost, materials, and structural efficiency



Advanced Design Tools modelling and simulation

Polynomial and Rational Functions

What you need to know

Understand Quadratic functions and their extrema.

Learn **how to Divide** polynomials

(Self Study)

Graphing Polynomial functions

(Self Study)

Evaluate polynomial and rational inequalities.

Understanding Polynomials

$$2x^2 + 4x^1 + 10x^0 = 0$$

expressions containing an unknown raised to a series of positive whole number exponents

$2x$ monomial	$2x - 1$ binomial	$x^2 + 2x - 1$ trinomial
One term	Two terms	Three terms

Degree of polynomials

$3x^3 + 5x^2 - 2x + 7 = 0$	highest power present is called the **degree** of the polynomial

$5x^2 - 2x + 7 = 0$

second degree polynomial or **quadratic** polynomial

$3x^3 + 5x^2 - 2x + 7 = 0$

third degree polynomial or **cubic** polynomial

$2x^5 + x^4 - 3x^3 + 5x^2 - 2x + 7 = 0$

beyond this we just say "n^{th} degree polynomial"

Polynomial functions: Formal definition

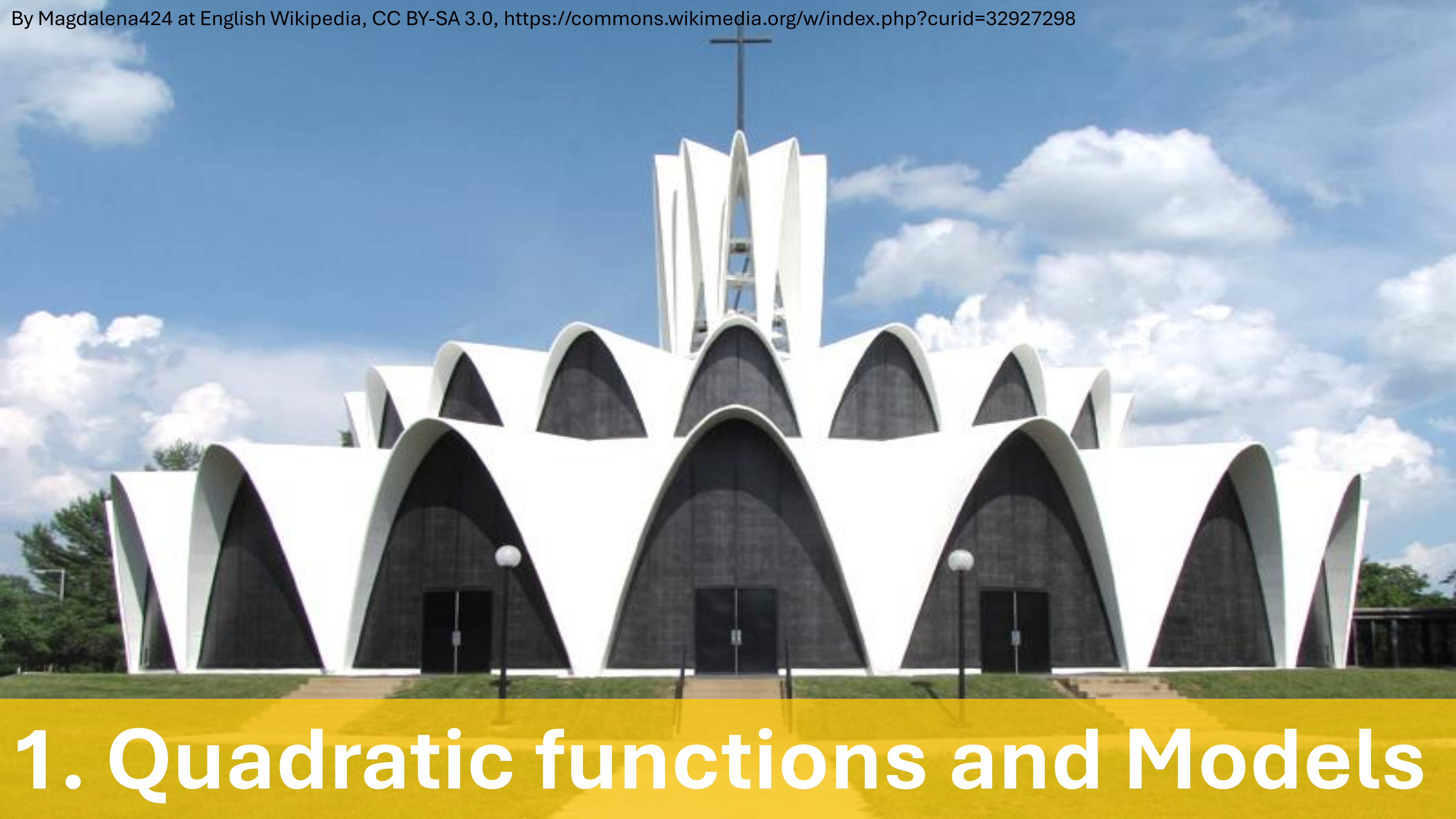
A polynomial function is a function that is defined by a polynomial expression. So a **polynomial function of degree *n*** is a function of the form

$$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \qquad a_n \neq 0$$

Convention: to write terms in decreasing order of exponents

$$\left.\begin{array}{c} 4x^2 + 7 = 0 \\ 4x^2 + 0x + 7 = 0 \end{array}\right\}$$

not all possible terms must be present



1. Quadratic functions and Models

Parabola and Modern Architecture

Garabit viaduct, France

Parabolic arches

- allow efficient use of materials because loads distributed horizontally
- Important for iron and steel structures to use the material with maximum efficiency in order to contain costs.

The parabola is especially seen in the style of the famous modern architect **Antonio Gaudí** who associated the parabola with natural, organic forms.

Parq Güell, Barcelona, Spain, 1900. Anton Gaudí

a. Quadratic Functions: General Form

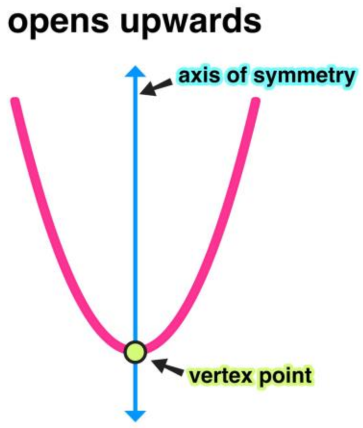
Quadratic Functions

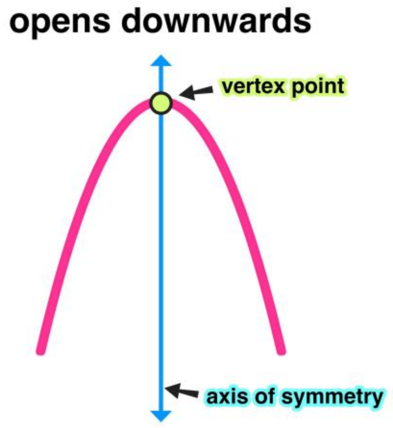
QUADRATIC FUNCTIONS

A **quadratic function** is a polynomial function of degree 2. So a quadratic function is a function of the form

$$f(x) = ax^2 + bx + c \qquad a \neq 0$$

What would happen if a was equal to zero?





The graph of a quadratic function is called a **parabola**

It has

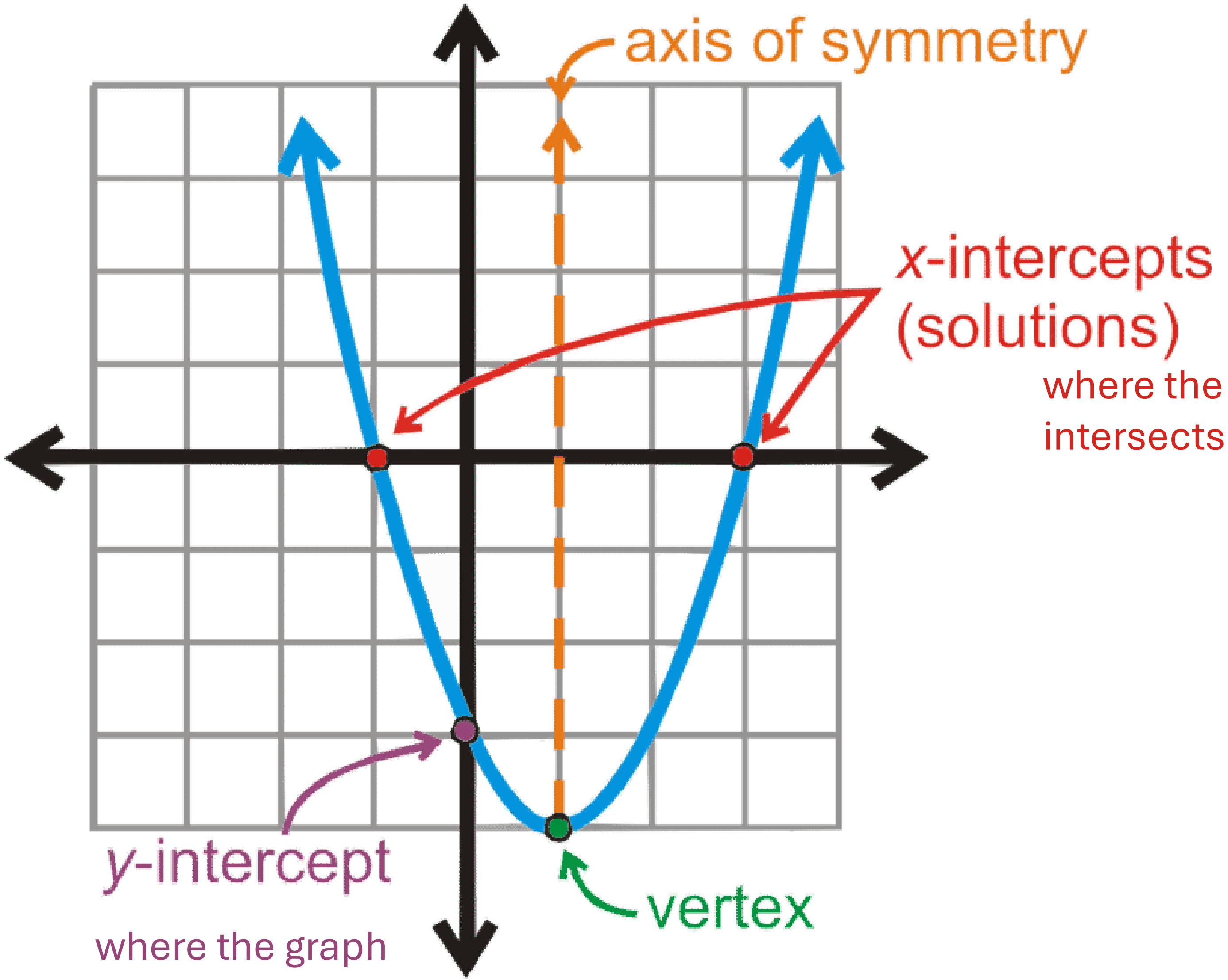
- 1 vertex
- 1 axis of symmetry

The parabolic form is also seen in the works of the Spanish Architect, **Santiago Calatrava**.

Here, Vaulting is formed with parabolic arches at BCE place, Toronto

BCE Place, Toronto, Canada, 1987-1992. Santiago Calatrava

Let’s get to know a parabola better



where the graph intersects the *x*-axis

where the graph intersects the *y*-axis

Examples of quadratic functions

The equation for the quadratic parent function is

$$y = x^2$$, where $x \neq 0$.

Here are a few quadratic functions:

$$ax^2 + bx + c$$

$$y = x^2 - 5$$

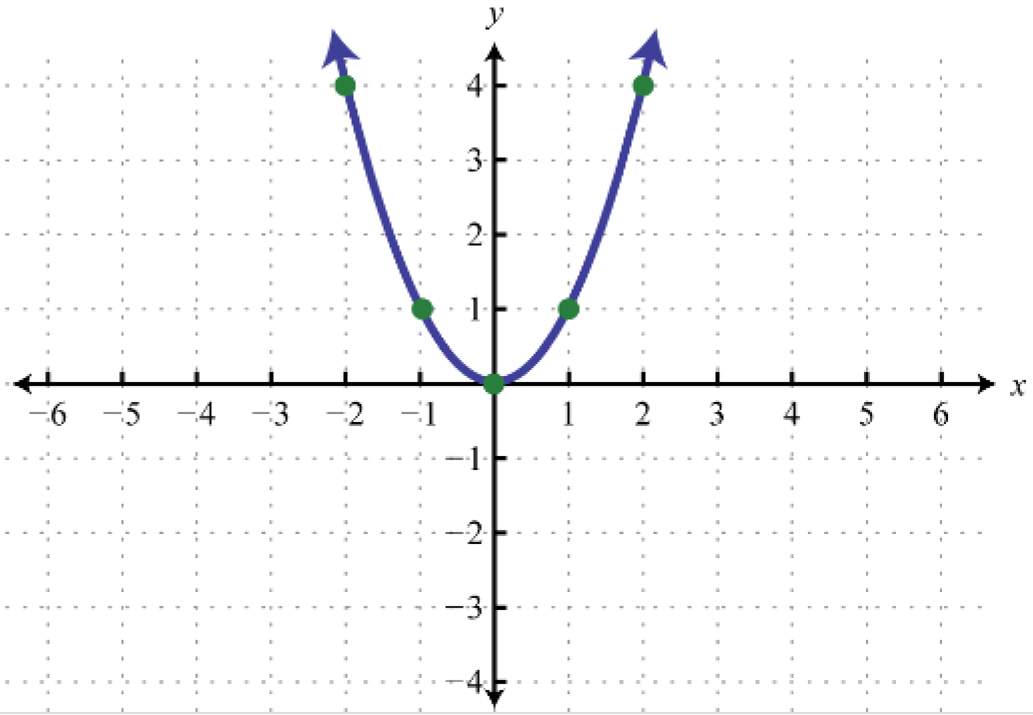
$$y = x^2 - 3x + 13$$

$$y = -x^2 + 5x + 3$$

A quadratic function and its graph

$f(x) = x^2$

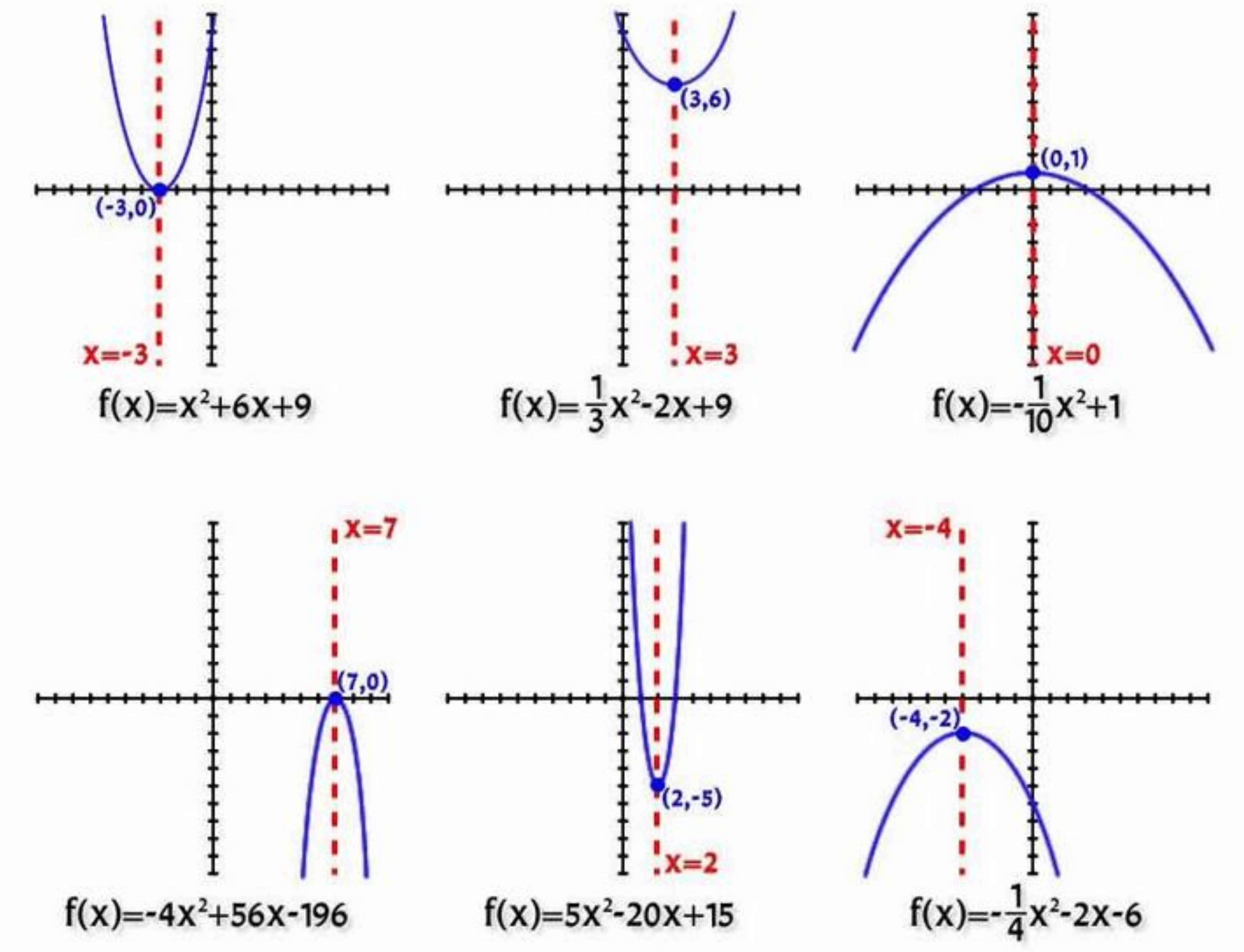
x	$f(x)$
-2	4
-1	1
0	0
1	1
2	4



Domain of this function consists of the set of **all real numbers $(-\infty,\infty)$**

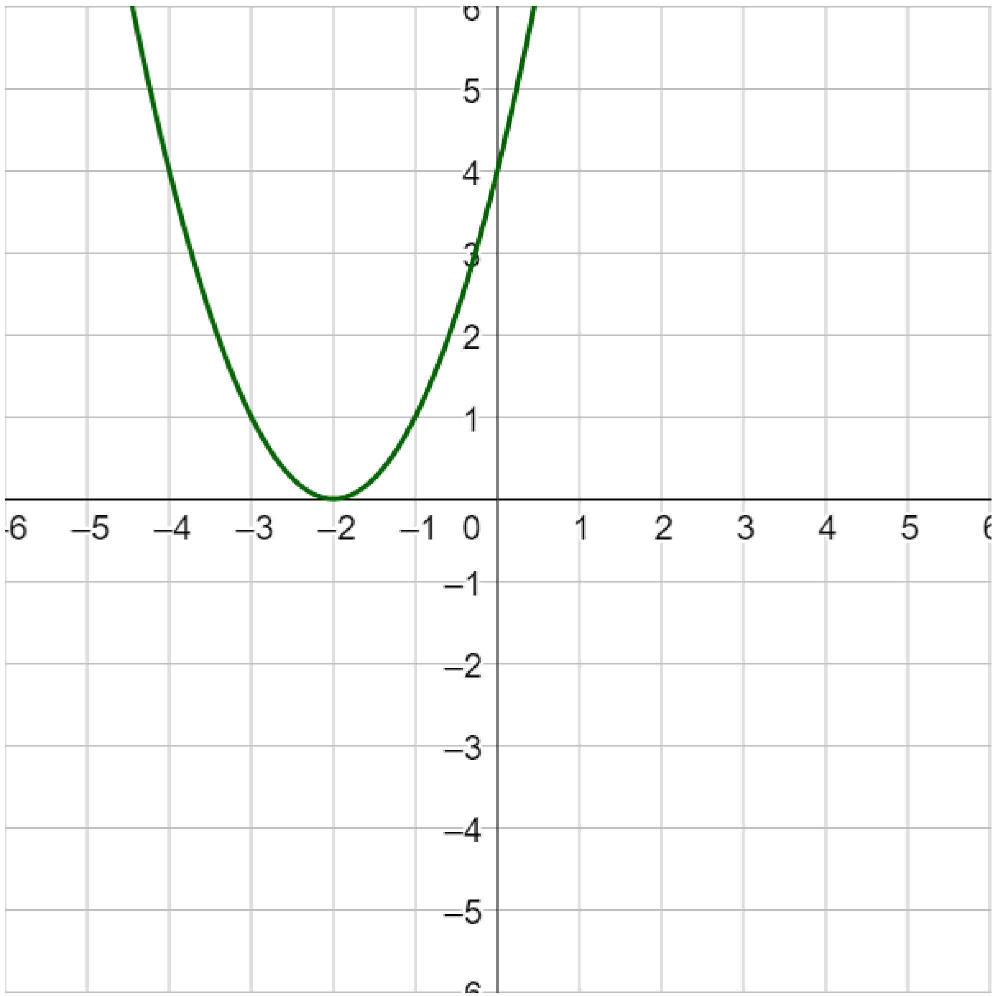
Range consists of the set of nonnegative numbers **$[0,\infty)$.**

Some quadratic functions and their graphs



A parabola can have
zero,
one,
or **two**
x-intercepts.

Identify parts of a parabola from a graph



Determine the vertex, axis of symmetry, zeros, and *y*-intercept of the parabola

ACTIVITY 1.1

1.1 Identify parts of a parabola from a graph

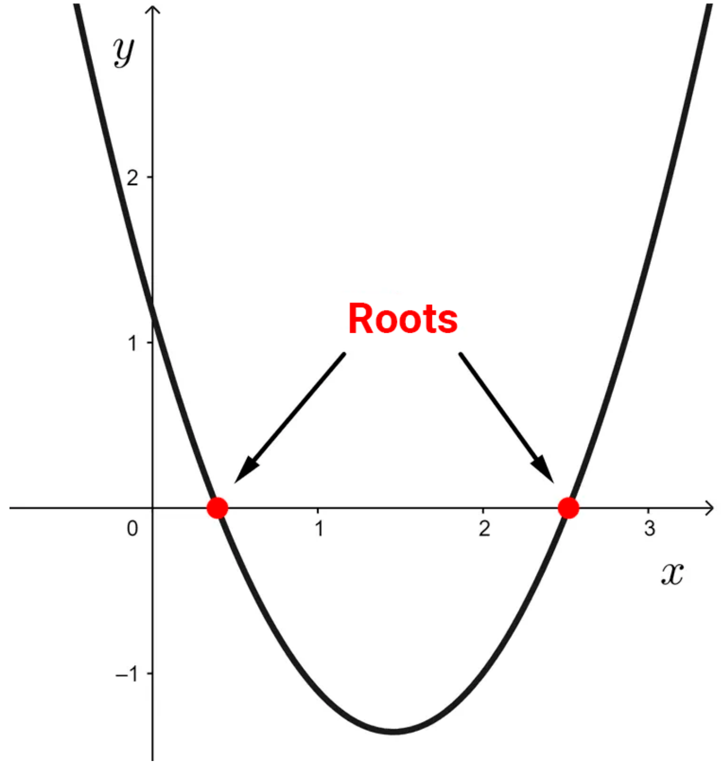
SOLUTION

What does "Solving a quadratic equation" mean?

Solving a quadratic equation means
finding the values of x
that satisfy the equation

$$ax^2+bx+c=0.$$

These values of x are referred to as the
solutions, **roots**, **zeros**, or
x-intercepts
of the quadratic equation.



Solving quadratics

Techniques to solve a quadratic equation

Graphing

Visual method

Factoring

simple quadratics where factors are easily identifiable

Completing the square

transforms the quadratic equation into a perfect square

Use the quadratic formula

when factoring is difficult

Solving quadratics: Completing the square

REVIEW FROM WEEK 1

Completing the square of an expression with multiple variables is a technique which manipulates the expression into **a perfect square plus some constant.**

$$x^2 + 6x = 2$$

This expression cannot be factored into a perfect square

$$x^2 + 6x + 9 = 2 + 9$$

We want to bring the polynomial to the form $(a + b)^2$

$$x^2 + 6x + 9 = 11$$

Half the coefficient of the term with degree 1 and **square** it;

$$(x + 3)^2 = 11$$

$6/2=3$

$3^2=9$

Add this constant to both sides of equation

Solving quadratics: Quadratic Formula

THE QUADRATIC FORMULA

The roots of the quadratic equation $ax^2 + bx + c = 0$, where $a \neq 0$, are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example of using the quadratic formula to find the solutions of a quadratic $3x^2 - 5x - 1 = 0$

In this quadratic equation $a = 3$, $b = -5$, and $c = -1$.

$b = -5$

$3x^2 - 5x - 1 = 0$

$a = 3$ $c = -1$

By the Quadratic Formula,

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(3)(-1)}}{2(3)} = \frac{5 \pm \sqrt{37}}{6}$$

EXAMPLE

Solving quadratic equations using the quadratic formula

(a) $x^2 + 5x - 6 = 0$

(b) $x^2 + 10x - 600 = 0$

(c) $4x^2 - 4x + 1 = 0$

ACTIVITY 1.2

1.2a Solving quadratic equations using quadratic formula

SOLUTION

1.2b Solving quadratic equations

SOLUTION

1.2c Solving quadratic equations

SOLUTION

The vertex: Max or Min value of a quadratic function (General form)

MAXIMUM OR MINIMUM VALUE OF A QUADRATIC FUNCTION

The maximum or minimum value of a quadratic function $f(x) = ax^2 + bx + c$ occurs at

$$x = -\frac{b}{2a}$$

If $a > 0$, then the **minimum value** is $f\left(-\frac{b}{2a}\right)$.

If $a < 0$, then the **maximum value** is $f\left(-\frac{b}{2a}\right)$.

If the parabola opens upward

then the vertex is the **lowest point** on the parabola.

If the parabola opens downward,

then the vertex is the **highest point** on the parabola.

Max and min values of Quadratic functions (Expressed in General form)

Find the maximum or minimum value of each quadratic function.

(a) $f(x) = x^2 + 4x$

(b) $g(x) = -2x^2 + 4x - 5$

SOLUTION

(a) This is a quadratic function with $a = 1$ and $b = 4$. Thus the maximum or minimum value occurs at

$$x = -\frac{b}{2a} = -\frac{4}{2 \cdot 1} = -2$$

Since $a > 0$, the function has the *minimum* value

$$f(-2) = (-2)^2 + 4(-2) = -4$$

(b) This is a quadratic function with $a = -2$ and $b = 4$. Thus the maximum or minimum value occurs at

$$x = -\frac{b}{2a} = -\frac{4}{2 \cdot (-2)} = 1$$

Since $a < 0$, the function has the *maximum* value

$$f(1) = -2(1)^2 + 4(1) - 5 = -3$$

EXAMPLE

Max and Min values using the vertex formula

Use the formula to determine the maximum or minimum values of the function

(a) $f(x) = 3 - 4x - x^2$

(b) $g(x) = 100x^2 - 1500x$

1.3 Max and Min values using the vertex formula

SOLUTION

b. Quadratic functions: Standard form

Standard form of a quadratic equation

"General form"

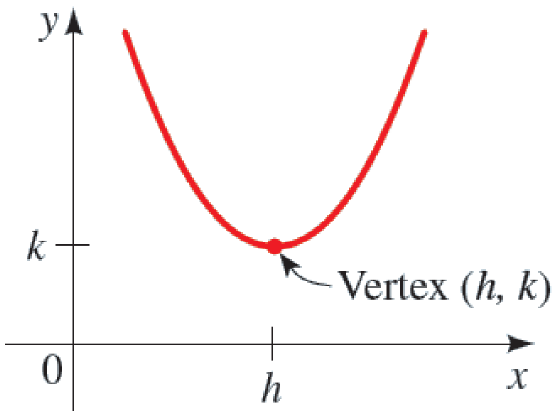
STANDARD FORM OF A QUADRATIC FUNCTION

A quadratic function $f(x) = ax^2 + bx + c$ can be expressed in the **standard form**

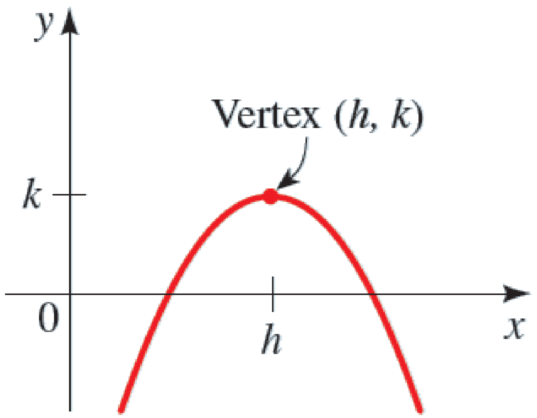
$$\boxed{f(x) = a(x - h)^2 + k}$$

"Standard form" or "Vertex Form"

by completing the square. The graph of f is a parabola with **vertex** (h, k); the parabola opens upward if $a > 0$ or downward if $a < 0$.



$f(x) = a(x - h)^2 + k, \ a > 0$

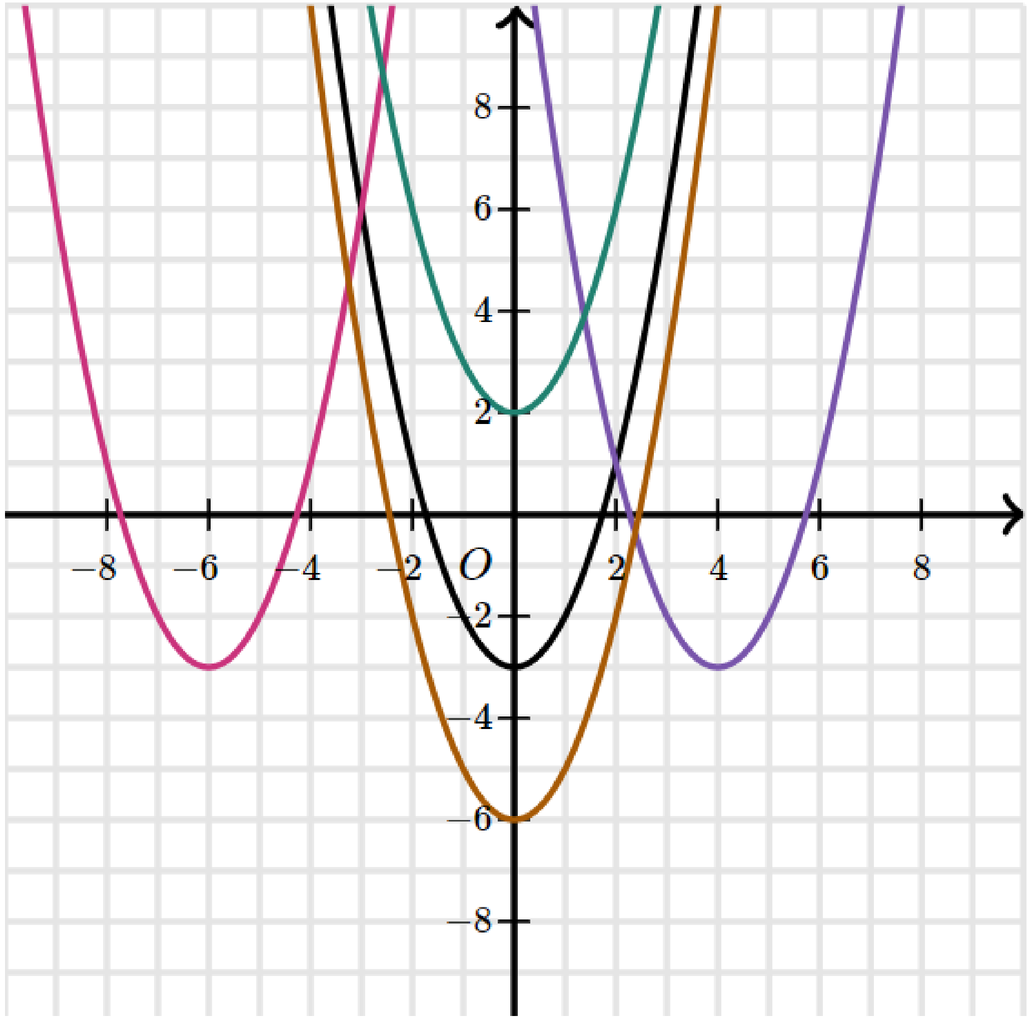


$f(x) = a(x - h)^2 + k, \ a < 0$

a is negative,
Graph looks like a frown

a is positive,
Graph looks like a happy face

Quadratic functions in the standard forms and their graphs



The graph below shows the graph of the quadratic function $f(x) = x^2 - 3$ alongside various translations

- The graph of $f(x-4) = (x-4)^2 - 3$ translates the graph of 4 units to the right.
- The graph of $f(x+6) = (x+6)^2 - 3$ translates the graph 6 units to the left.
- The graph of $f(x) + 5 = x^2 + 2$ translates the graph 5 units up.
- The graph of $f(x) - 3 = x^2 - 6$ translates the graph 3 units down.

Standard form of a quadratic equation

Let $f(x) = 2x^2 - 12x + 13$.

(a) Express f in standard form.

(b) Find the vertex and x- and y-intercepts of f.

(c) Sketch a graph of f.

(d) Find the domain and range of f.

EXAMPLE

Solution (a): Standard form of a quadratic equation

(a) Use ‘complete the squares’ technique to express the function in its standard form!

(a) Since the coefficient of x^2 is not 1, we must factor this coefficient from the terms involving x before we complete the square.

$$\begin{aligned} f(x) &= 2x^2 - 12x + 13 \\ &= 2(x^2 - 6x) + 13 && \text{Factor 2 from the } x\text{-terms} \\ &= 2(x^2 - 6x + 9) + 13 - 2\cdot 9 && \text{Complete the square: Add 9 inside parentheses, subtract } 2\cdot 9 \text{ outside} \\ &= 2(x-3)^2 - 5 && \text{Factor and simplify} \end{aligned}$$

The standard form is $f(x) = 2(x - 3)^2 - 5$.

EXAMPLE

Solution (b): Standard form of a quadratic equation

(b) Read the vertex from the standard form of the equation

The standard form is $f(x) = 2(x - 3)^2 - 5$.

From the standard form of f we can see that the vertex of f is $(3, -5)$

The y-intercept is $f(0) = 13$.
To find x-intercepts, set the function equal to zero.
To find x-intercepts (roots of the equation), we can use quadratic formula:

$$0 = 2x^2 - 12x + 13$$ Set $f(x) = 0$

$$x = \frac{12 \pm \sqrt{144 - 4 \cdot 2 \cdot 13}}{4}$$ Solve for x using the Quadratic Formula

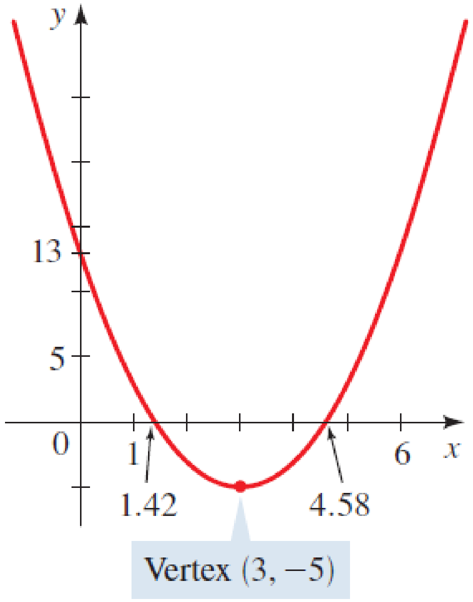
$$x = \frac{6 \pm \sqrt{10}}{2}$$ Simplify

EXAMPLE

Solution (c), (d): Standard form of a quadratic equation

Sketch the graph using principles of transformations we learnt previously!

The standard form tells us that we get the graph of f by taking the parabola $y = x^2$, , shifting it to the right 3 units, stretching it vertically by a factor of 2, and moving it downward 5 units.



The domain of f is the set of all real numbers $(-\infty, \infty)$. From the graph we see that the range of f is $[-5, \infty)$.

EXAMPLE

Maximum and minimum values of Quadratic Functions

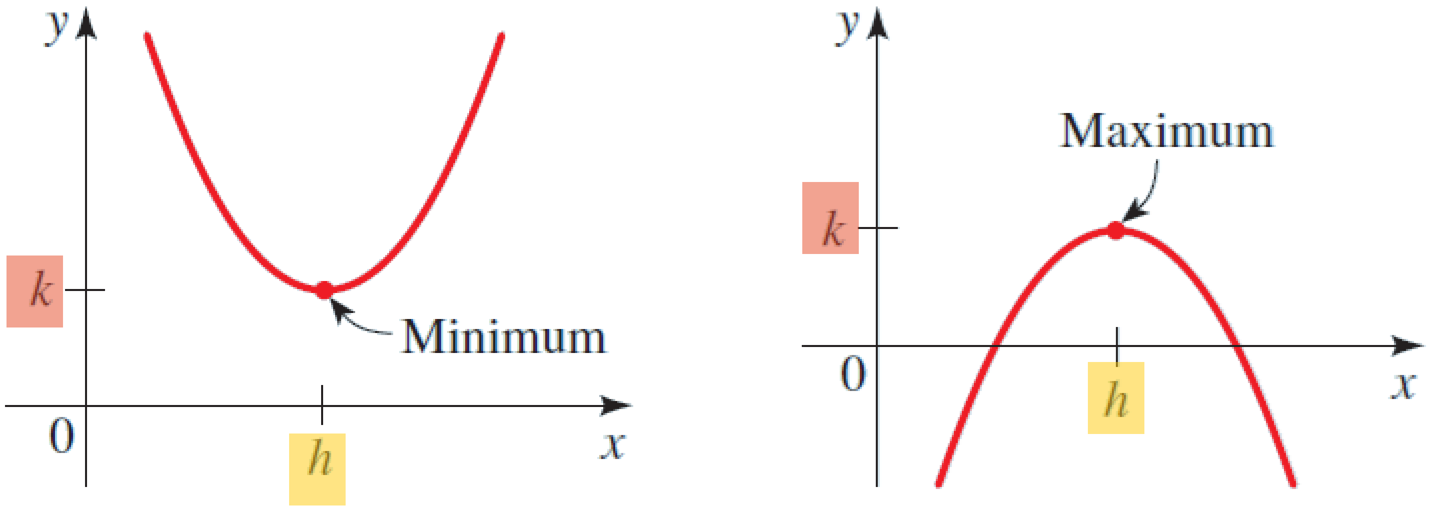
MAXIMUM OR MINIMUM VALUE OF A QUADRATIC FUNCTION

Let f be a quadratic function with standard form $f(x) = \boxed{a(x - h)^2 + k.}$ The maximum or minimum value of f occurs at $x = h$.

If $a > 0$, then the **minimum value** of f is $f(h) = k$.

If $a < 0$, then the **maximum value** of f is $f(h) = k$.

Known as 'vertex form' as you can read the vertex directly!



$f(x) = a(x - h)^2 + k,\ a > 0$ $\qquad$ $f(x) = a(x - h)^2 + k,\ a < 0$

Quiz: Standard form of a parabola

If the function $f(x) = (x - 3)^2 - 11$ is graphed in the xy-plane, what are the coordinates of the vertex?

Choose 1 answer:

- (A) $(-3, -11)$
- (B) $(-3, 11)$
- (C) $(3, -11)$
- (D) $(3, 11)$

ACTIVITY

Quiz: Solution

SOLUTION

Min Value of a Quadratic function (Standard form)

Consider the quadratic function $f(x) = 5x^2 - 30x + 49$

(a) Express f in standard form.

(b) Sketch a graph of f.

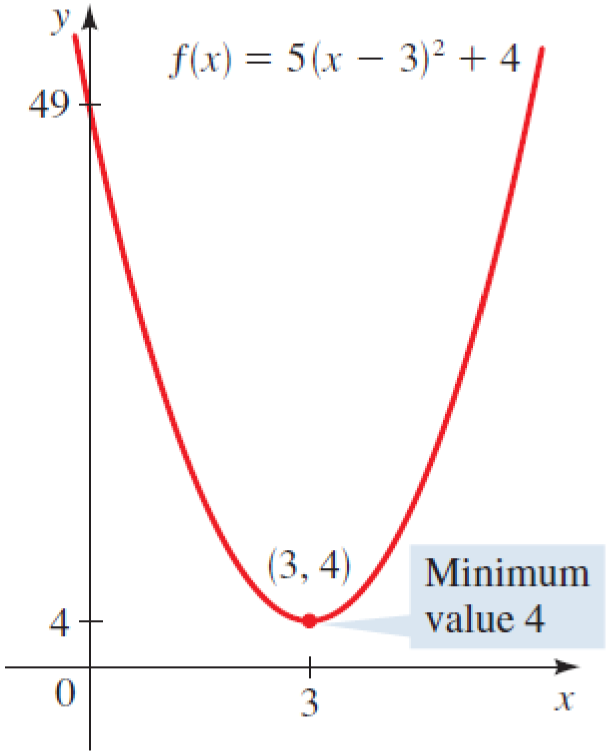
(c) Find the minimum value of f.

SOLUTION

(a) To express this quadratic function in standard form, we complete the square.

$$\begin{aligned} f(x) &= 5x^2 - 30x + 49 \\ &= 5(x^2 - 6x) + 49 && \text{Factor 5 from the } x\text{-terms} \\ &= 5(x^2 - 6x + 9) + 49 - 5 \cdot 9 && \text{Complete the square: Add 9 inside parentheses, subtract } 5 \cdot 9 \text{ outside} \\ &= 5(x-3)^2 + 4 && \text{Factor and simplify} \end{aligned}$$

(b) The graph is a parabola that has its vertex at $(3, 4)$ and opens upward, as



(c) Since the coefficient of x^2 is positive, f has a minimum value. The minimum value is $f(3) = 4$.

EXAMPLE

Example: Max Value of a Quadratic function (Standard form)

Consider the quadratic function $f(x) = -x^2 + x + 2$.

(a) Express f in standard form.

(b) Sketch a graph of f.

(c) Find the maximum value of f.

EXAMPLE

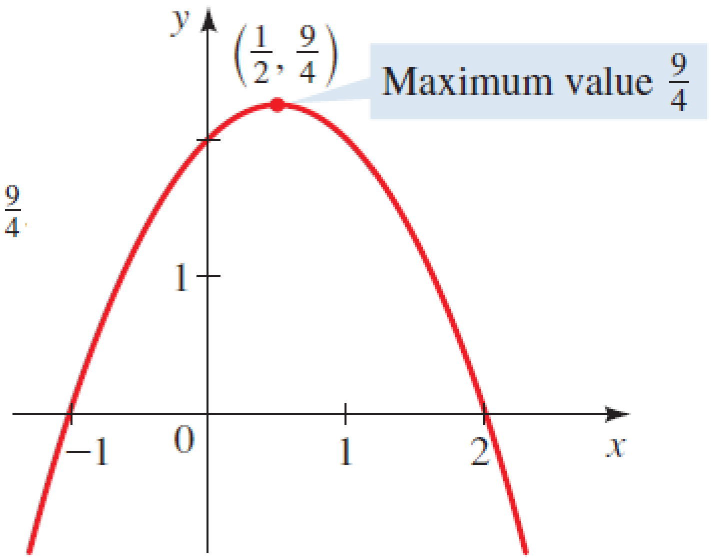
Solution: Max value of a quadratic function (Standard form)

(a) To express this quadratic function in standard form, we complete the square.

$$\begin{aligned} f(x) &= -x^2 + x + 2 \\ &= -(x^2 - x) + 2 && \text{Factor } -1 \text{ from the } x\text{-terms} \\ &= -\left(x^2 - x + \tfrac{1}{4}\right) + 2 - (-1)\tfrac{1}{4} && \text{Complete the square: Add } \tfrac{1}{4} \text{ inside parentheses, subtract } (-1)\tfrac{1}{4} \text{ outside} \\ &= -\left(x - \tfrac{1}{2}\right)^2 + \tfrac{9}{4} && \text{Factor and simplify} \end{aligned}$$

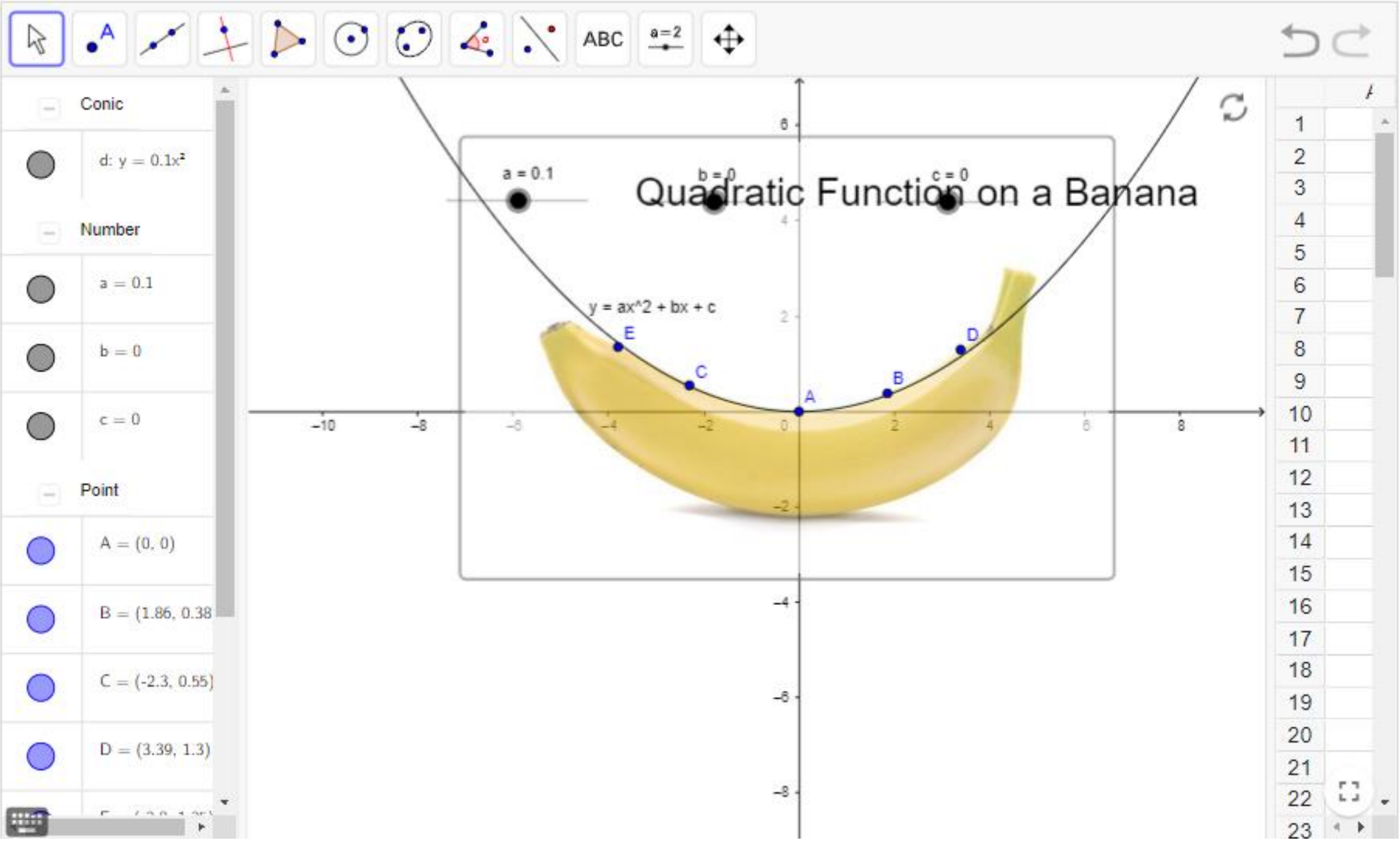
(b) From the standard form we see that the graph is a parabola that opens downward and has vertex $\left(\frac{1}{2}, \frac{9}{4}\right)$.

(c) Since the coefficient of x^2 is negative, f has a maximum value, which is $f\left(\frac{1}{2}\right) = \frac{9}{4}$.



EXAMPLE

Quadratic function on a banana



https://www.geogebra.org/m/C9zPphwt

Max and Min values (Standard form)

A quadratic function f is given. **(i)** Express f in standard form. **(ii)** Sketch a graph of f. **(iii)** Find the maximum or minimum value of f.

(a) $f(x) = x^2 - 8x + 8$

(b) $f(x) = 1 - 6x - x^2$

ACTIVITY 1.4

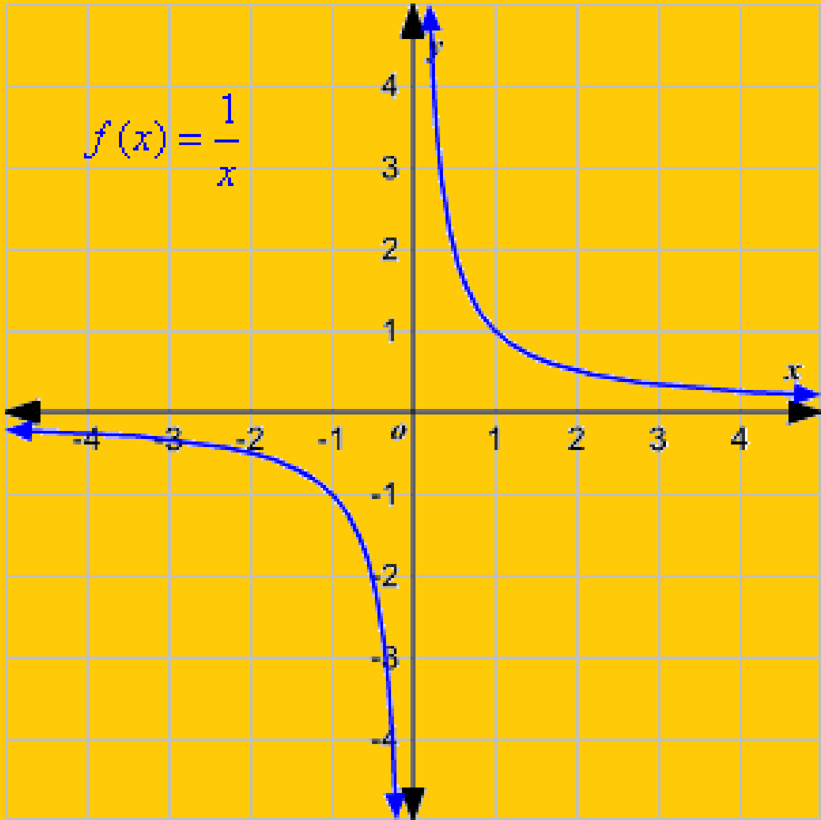
1.4 a: Max and Min values (Standard form)

SOLUTION

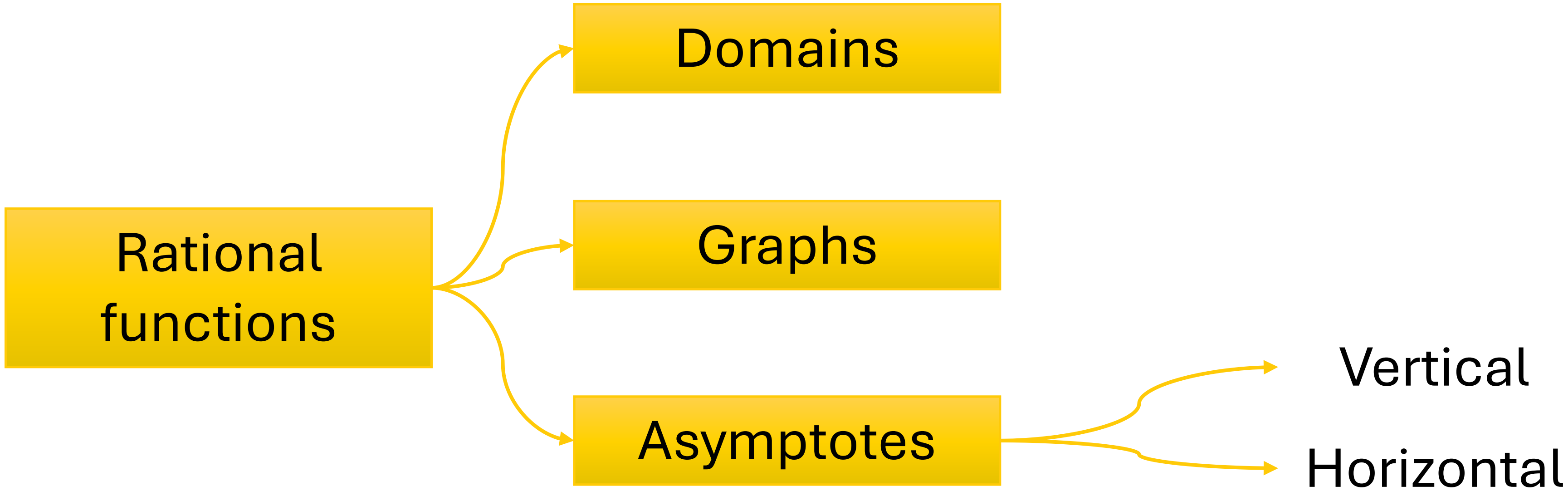
1.4 b: Max and Min values (Standard form)

SOLUTION

2. Rational functions



Overview


Rational functions
Domains
Graphs
Asymptotes
Vertical
Horizontal

Rational functions

rational means "ratio" or can be written as a fraction.

Rational functions are functions written as **fractions of polynomial functions** in the form

$$f(x) = \frac{P(x)}{Q(x)}$$

Variable in the denominator

where $P(x)$ and $Q(x)$ are polynomial functions.
(Assuming there is no factor in common)

EXAMPLE

$$f(x) = \frac{x^2 - 3}{x + 5}$$

Denominator cannot be zero

$$x \neq -5$$

→The ***domain*** of a rational function consists of

all real numbers *x* **except those for which the denominator is zero**

→graph of a rational function will have **breaks** or **holes** at those x**-values where it is undefined.**

Rational functions: Where are these used?

Example: An average cost function

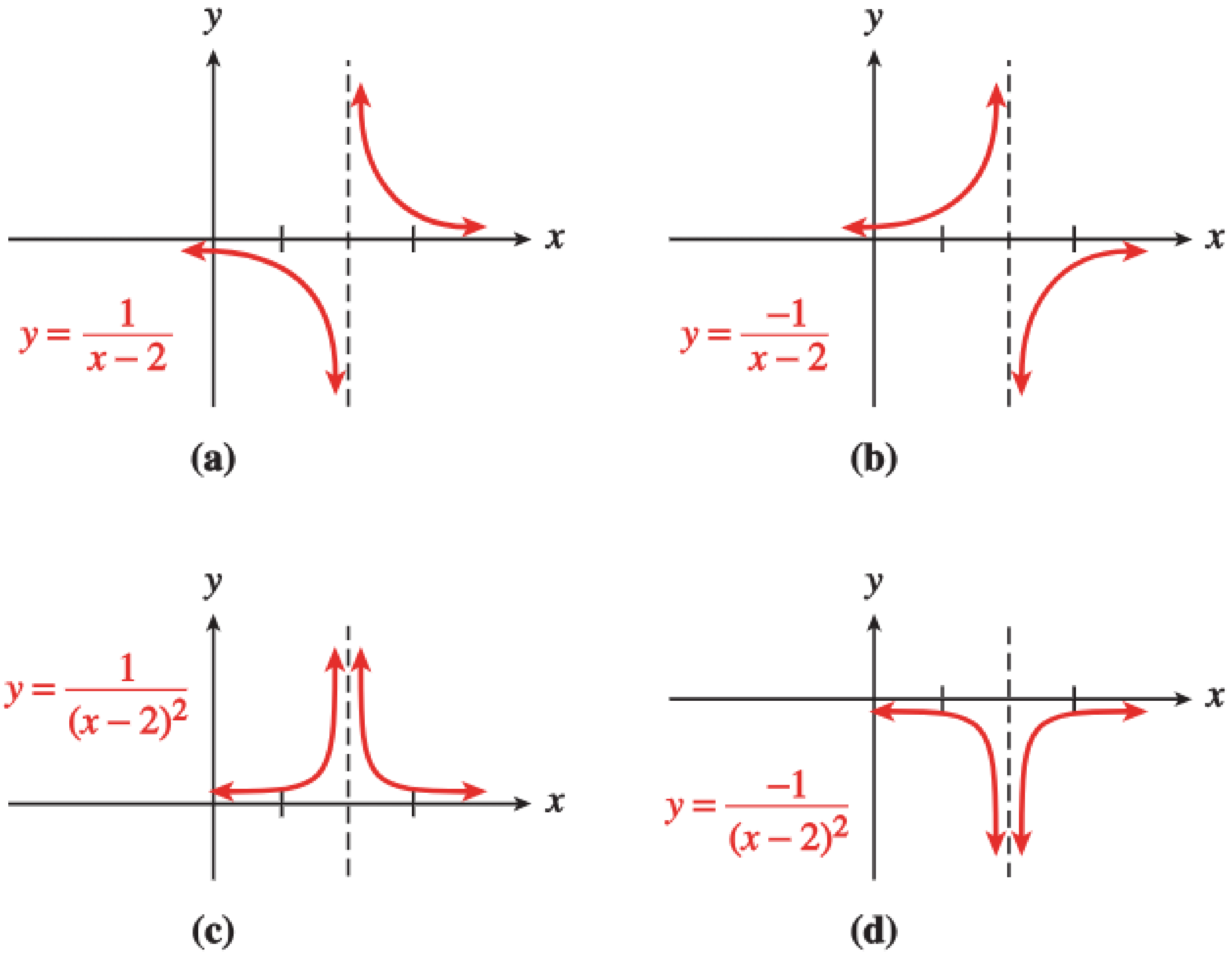
$$f(x) = \frac{125x + 2000}{x}$$

Cost to manufacture x items

Average cost for producing **x** items is found by dividing the cost function by the number of items, **x**

Variable in the denominator
→ Rational function

Graphs of rational functions: More Examples



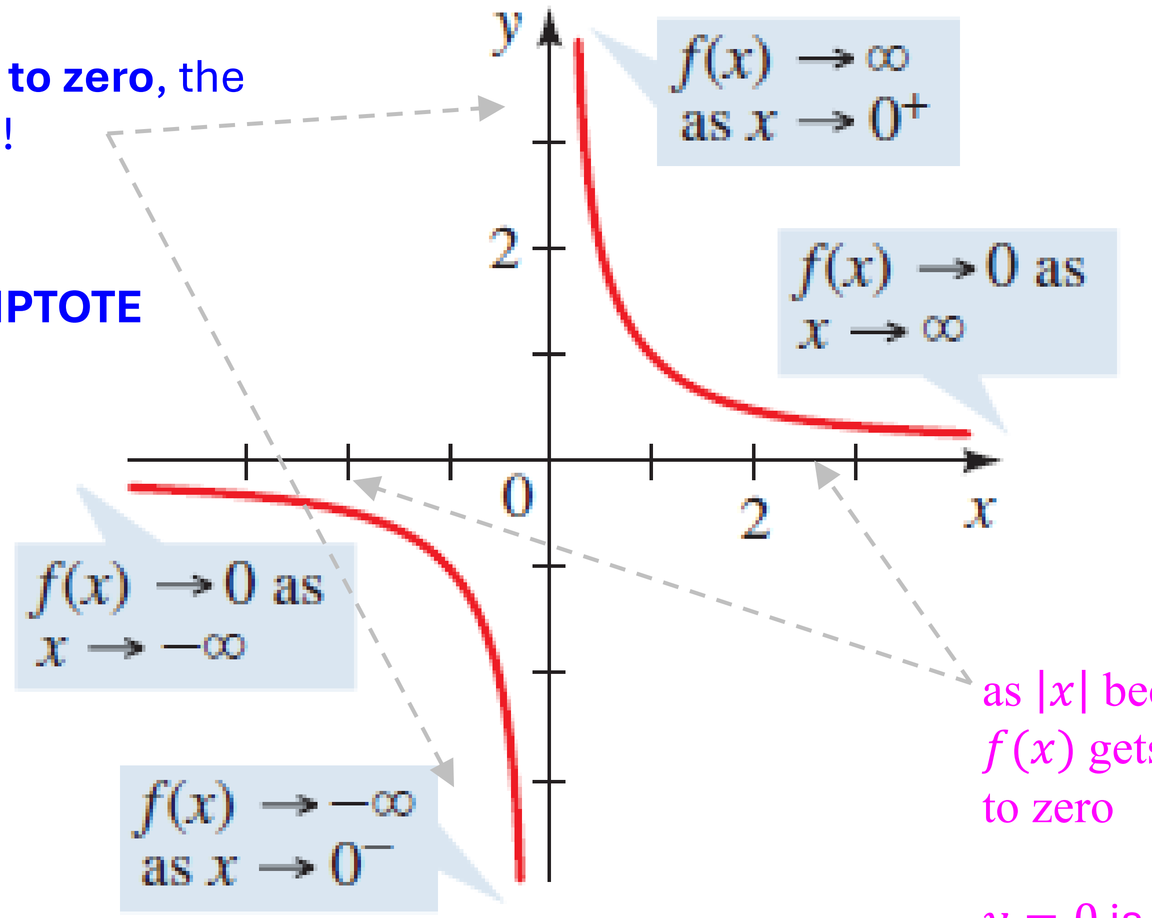
graphs of rational functions are characterized by **asymptotes**

Graphs of rational functions: Example

$f(x) = \frac{1}{x}$ The function f is not defined for $x = 0$

the closer x **gets to zero**, the larger $|f(x)|$ gets!

$x = 0$ is a
VERTICAL ASYMPTOTE



as $|x|$ becomes large, the value of $f(x)$ gets closer and closer to zero

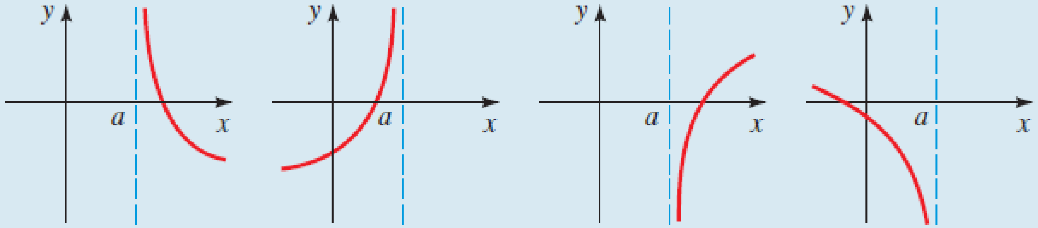
$y = 0$ is a
HORIZONTAL ASYMPTOTE

Vertical and horizontal asymptotes

Asymptotes are lines that the curve approaches at the edges of the coordinate plane.

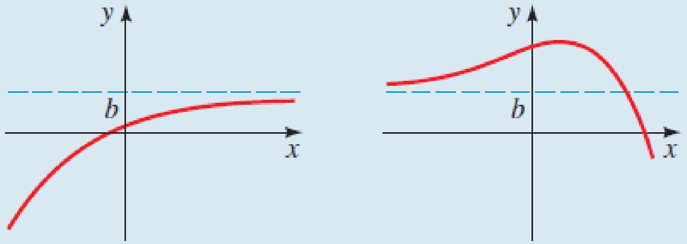
DEFINITION OF VERTICAL AND HORIZONTAL ASYMPTOTES

1. The line $x = a$ is a **vertical asymptote** of the function $y = f(x)$ if y approaches $\pm\infty$ as x approaches a from the right or left.



$y \to \infty$ as $x \to a^+$ $\quad$ $y \to \infty$ as $x \to a^-$ $\quad$ $y \to -\infty$ as $x \to a^+$ $\quad$ $y \to -\infty$ as $x \to a^-$

2. The line $y = b$ is a **horizontal asymptote** of the function $y = f(x)$ if y approaches b as x approaches $\pm\infty$.



$y \to b$ as $x \to \infty$ $\quad$ $y \to b$ as $x \to -\infty$

Vertical asymptotes occur where the **denominator of a rational function approaches zero**.

Horizontal asymptotes occur when the function approaches a constant value as ***x*-values get very large in the positive or negative direction**.

How to find the domain of a rational function

The domain of a rational function cannot include a value that makes the denominator equal zero because that causes the function to be undefined.

Domain of a Rational Function

The domain of a rational function is all real numbers **except those that cause the denominator to equal zero.**

Find the Domain of a Rational Function

1. Set the denominator equal to zero.
2. Solve to find the *x*-values that cause the denominator to equal zero.
3. The domain is all real numbers except those found in step 2.

Find the domain of a rational function

Find the domain of $f(x) = \frac{x-2}{x^2-4}$

Set the denominator equal to zero and solve for *x*.

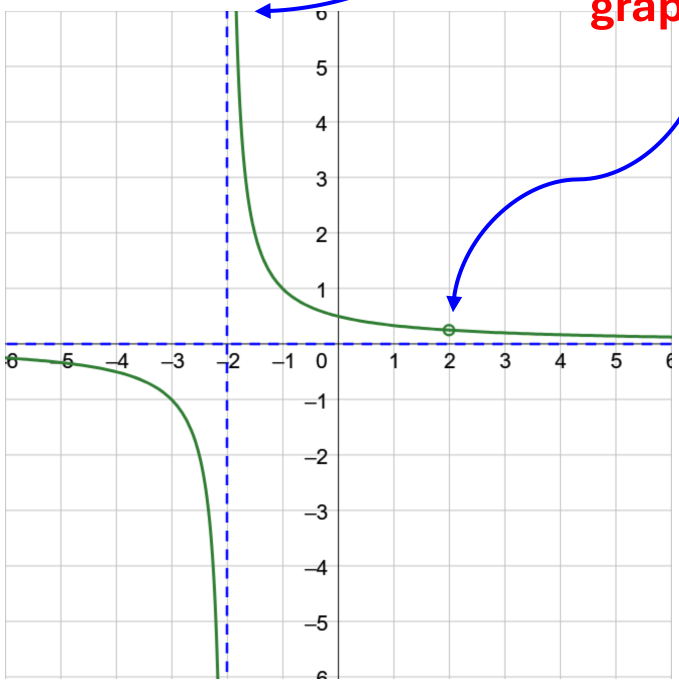
$x^2 - 4 = 0$
$x^2 = 4$
$x = \pm 2$

The domain of the function is all real numbers except $x = \pm 2$

Function is not defined when $x = \pm 2$

Holes occur where both the numerator and the denominator are zero at the same point

vertical asymptote at x = –2 and **a hole in the graph at x = 2**



EXAMPLE

How to find identify vertical asymptotes of a rational function

Vertical asymptotes come from the **factors of the denominator that are not in common with a factor of the numerator. The vertical asymptotes occur where those factors equal zero**

Identify Vertical Asymptotes of a Rational Function

1. Factor the numerator and denominator.
2. Simplify by canceling common factors in the numerator and the denominator.
3. Set the simplified denominator equal to zero and solve for x.

Identifying Vertical asymptotes

Find the vertical asymptotes of the graph of $g(x) = \frac{x-2}{x^2-4x+3}$

ACTIVITY 2.1

4.1 Identifying vertical asymptotes

SOLUTION

How to determine horizontal asymptotes

Vertical asymptotes describe the behavior of a graph as **the output approaches ∞ or −∞**.
Horizontal asymptotes describe the behavior of a graph as **the input approaches ∞ or −∞.**

To help us find this value, **we divide both numerator and denominator by the highest power of x that appears in the expression**.

Find the horizontal asymptote of

$$r(x) = \frac{2x^2 - 4x + 5}{x^2 - 2x + 1}$$

$$y = \frac{2x^2 - 4x + 5}{x^2 - 2x + 1} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} = \frac{2 - \frac{4}{x} + \frac{5}{x^2}}{1 - \frac{2}{x} + \frac{1}{x^2}}$$

These terms approach 0

$$y = \frac{2 - \frac{4}{x} + \frac{5}{x^2}}{1 - \frac{2}{x} + \frac{1}{x^2}} \longrightarrow \frac{2 - 0 + 0}{1 - 0 + 0} = 2$$

These terms approach 0

The fractional expressions $\frac{4}{x}, \frac{5}{x^2}, \frac{2}{x}$, and $\frac{1}{x^2}$ all approach 0 as $x \to \pm\infty$

Thus the horizontal asymptote is the line $y = 2$

EXAMPLE

Summary: Finding asymptotes

FINDING ASYMPTOTES OF RATIONAL FUNCTIONS

Let r be the rational function

$$r(x) = \frac{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0}$$

1. The vertical asymptotes of r are the lines $x = a$, where a is a zero of the denominator.
2. (a) If $n < m$, then r has horizontal asymptote $y = 0$.

   (b) If $n = m$, then r has horizontal asymptote $y = \dfrac{a_n}{b_m}$.

   (c) If $n > m$, then r has no horizontal asymptote.

NOTE
In 1), Numerator and Denominator must have no common factors

Identifying Vertical and horizontal asymptotes

(a) $r(x) = \dfrac{2x - 3}{x^2 - 1}$

(b) $s(x) = \dfrac{8x^2 + 1}{4x^2 + 2x - 6}$

ACTIVITY 2.2

4.2a Identifying Vertical and horizontal asymptotes

SOLUTION

4.2b Identifying Vertical and horizontal asymptotes

SOLUTION